DECISION RECORD

Learning without barriers

Finalized product direction from the supplied LOS discussion. Prepared for Aaryan Kasaudhan.

Mission: Let anyone begin learning from a phone, including people without a school, teacher or linked parent. Extra support should strengthen access, never become a condition for it.

Area	Decision
Access	Self-registration; independent learning or optional class membership. No admin approval required to begin learning.
Support network	Teachers self-register. Parents use separate, optionally linked accounts. Learner history belongs to the learner.
Parent reports	Dashboard reports remain free. Email summaries are a separately purchased, optional add-on.
Learning signals	Combine self-report, activity/tests and understanding checks. Parent/teacher feedback is an optional additional input.
Privacy	Camera processing stays on-device. Raw images/video do not leave the device. Only necessary summaries sync with clear consent.
Architecture	Shared backend APIs, MongoDB and Firebase Authentication; offline lessons and practice with later progress sync.
AI economics	Zero initial AI-hosting budget; limited free-tier AI, optional local AI, no automatic paid fallback. Progress tracking stays free.
Paid access	Free, Pro and Ultra; monthly AI credits plus optional top-ups. Premium model use is metered.
Future scope	Personalized explanatory videos, higher-quality 3D visuals and extra teacher capabilities belong to a later phase.

How to read this document: These are agreed product decisions, not a claim that the repository already implements them. Engineering guidance is identified where it extends the discussion. No previously asked product question is left awaiting an answer.

01 / ACCOUNTS AND INCLUSION

One learner, many paths

The learning account stays independent of any school, teacher, class or paid subscription.

Learner onboarding

Anyone can create an account and start with **Learn independently** or **Join a class**. A teacher, parent or admin is not required for independent learning. The experience must work for phone-only learners and those outside formal schooling.

Optional class membership

A learner can join a teacher or class through an invitation/linking workflow and can leave later. Their account, personal progress and learning history remain available after leaving. Support more than one teacher when needed.

Teacher and parent accounts

Teachers self-register and manage their teaching relationships. Parents/guardians have their own accounts and optionally link to the learner; they do not share the learner login. A learner can have more than one guardian. A missing parent link must not restrict learning.

Admin responsibilities

Admins manage platform operations, accounts and relationship problems. Learning access must not wait for an admin to create or approve every learner account. Relationship verification is separate from permission to begin learning.

Role	Primary scope
Learner	Own lessons, practice, focus tools, learning history and optional support links.
Teacher / mentor	Courses, assignments, linked students and permitted learning insights.
Parent / guardian	Reports and support information for linked learners; optional email reporting.
Admin	Account administration, link management and platform operations.

Implementation guidance: Store teacher-student and guardian-student relationships as separate ID-based records, rather than a single teacher field in a learner profile. Validate link status and permissions before exposing another person's information.

02 / PERSONALIZATION AND PRIVACY

Useful signals, clear limits

Learning support is based on estimates and evidence of understanding, not a definitive measurement of mental state.

Input	Use and boundary
Self-report	Learner reports perceived focus, difficulty or overload. Keep check-ins simple.
Study activity and tests	Use learning activity, quiz results and progress to estimate where support is useful.
Understanding checks	Ask for an explanation in the learner's own words or a concrete example. Allow short written responses and an optional voice-answer route.
Parent / teacher feedback	Optional extra context where a linked adult can provide it. Its absence must never count as poor performance.

Camera and microphone controls

Both are optional, separately controlled and default off. The learner enables them for a session and can stop them at any time. Learning remains fully available with both off. Show a visible indicator while capture is active.

On-device image processing

Process camera images on the learner's device. Raw camera images and video must not be uploaded or stored in MongoDB. Camera signals are limited to basic session cues such as face presence and lighting; do not present them as proof of attention, emotion or cognitive load.

Microphone use

Use the mic for a learner-initiated voice answer, not continuous background listening. The discussion did not authorize routine audio retention; do not assume that image-processing consent permits uploading voice recordings.

Summary sync and visibility

Only necessary derived summaries sync to the backend with clear consent. Internal algorithms and optional feedback workflows need not be promoted on the public About page or feature list. Keep meaningful data-use and recording controls visible to the affected user; private implementation details do not require hidden data collection.

Equal access rule: Independent and class-linked learners both receive understanding checks and adaptive support. Optional adult feedback adds context without becoming a requirement.

03 / DATA AND OFFLINE USE

Shared backend, local continuity

Desktop and mobile use the same account, APIs and learning records, with equivalent access to plan entitlements.

Agreed technology direction

Keep **MongoDB** as the structured NoSQL database and **Firebase Authentication** for sign-in. Frontends communicate with a shared backend API; they do not connect directly to the database. Apply schemas, validation and duplicate-link prevention even though MongoDB is NoSQL.

Data group	Storage direction
Profiles and links	Learner/teacher/guardian profiles and separate relationship records linked by IDs.
Learning content	Courses, modules, lessons, notes, tests, homework and tasks.
Learning history	Quiz results, completions, progress and necessary consented session summaries.
Consent	Record which optional capture/sync functions the user has allowed.
Raw camera media	On-device processing only; no raw image/video upload or database storage.

Offline learning

Downloaded lessons, notes and saved practice quizzes remain usable without internet. Record progress locally and sync it after connectivity returns. A learner must be able to continue useful study when cloud AI is unavailable.

Operations that need internet

Cloud AI requires a connection. Final assignment submission to a teacher also requires internet. The product should distinguish local work saved on the device from a submission that the server has actually received.

Device continuity

After successful sync, the learner can continue across desktop and mobile. Subscription entitlements and premium model availability should apply consistently across both, subject to device capability for local AI.

Implementation guidance: Queue local progress updates with stable event IDs; retry safely without double-counting. Show sync/submission status. Enforce relationship permissions on the server and check Firebase identity on protected API calls.

04 / AI AND ZERO-BUDGET LAUNCH

Six roles, shared inference

Keep the specialized agent responsibilities while reducing duplicate model infrastructure and unnecessary calls.

Free launch architecture

Use a single shared free-tier cloud model behind six role-specific instruction sets: focus content, task hints, learning insights, mentor interventions, task/test generation and supportive SOS chat. The selected pilot direction from the discussion was **Groq with gpt-oss-20b**.

Model selection boundary

The provider/model names above record the discussion choice, not a fresh availability or quality guarantee. Account eligibility, live model availability and quotas must be verified at integration time. Do not hard-code the previously mentioned free-tier numbers as a permanent product promise.

Reduce inference demand

Provide the relevant lesson context with requests. Reuse appropriate common explanations and saved quizzes. Keep marks, completion counts, study time and progress calculations in ordinary application code. These deterministic operations should not consume AI credits.

Always-available free progress tracker

Progress tracking must continue without premium AI. Use normal code and locally saved activity to calculate the basic tracker, then sync it when online. A small local model can be considered later for optional narrative summaries; it is not required to keep progress working.

Quota and failure behavior

Disable automatic paid-provider fallback for the free launch. If the free quota is exhausted or the provider fails, show that live AI is temporarily unavailable. Lessons, notes, saved practice and the progress tracker continue. Do not substitute mock answers while presenting them as live AI output.

Optional local AI

Qwen3.5-4B was discussed as an optional desktop candidate, subject to hardware suitability and quality checks. Local AI must not be compulsory for budget phones. Open weights and free hosted usage do not imply free hardware, hosting or unlimited capacity.

Integration quality gate: Evaluate Hindi/English explanations, maths, lesson-grounded answers and quiz correctness before enabling a replacement model. Keep supportive SOS responses within a clearly described wellbeing-support role.

05 / PLANS AND REPORTING

Free core, metered upgrades

The initial AI-hosting budget is zero. Paid options can fund additional inference without restricting basic learning.

Plan / add-on	Agreed direction
Free	Core learning, free progress tracking and limited basic AI. Free linked-parent dashboard reports. No unlimited cloud AI promise.
Pro	Monthly AI allowance, selected premium models and more usage than Free. Optional additional credit purchases.
Ultra	A larger allowance and greater access to costly premium models. Usage remains capped and metered.
Parent email add-on	Optional paid weekly email summaries for parents who choose and purchase them. Dashboard reports remain free.

Credits rather than equal token counts

Use a common credit allowance for the user-facing plan. Charge different credit amounts for different models according to their underlying cost, rather than treating one token from every model as financially equivalent. Show the cost of a request clearly enough for users to manage their allowance.

Monthly subscription plus optional top-ups

The accepted commercial structure is a recurring monthly allowance with optional top-up credits. Do not silently purchase credits or route a free user to a paid provider when free capacity runs out.

Premium model access

Paid plans can offer selected models from providers such as Gemini, OpenAI and Claude. Model labels mentioned conversationally are not a locked catalog. Keep the model roster configurable and verify supported names at integration time. Paid access should carry across desktop and mobile.

Student affordability

Preserve the objective of accessible learning. Student discounts can be offered; no exact discount, rupee price, monthly credit quantity or expiry/rollover rule was agreed. This document therefore records the commercial structure without inventing financial terms.

Commercial boundary: These are product-direction decisions. They do not activate billing, send parent emails, purchase inference or establish a finalized price list.

06 / ROADMAP AND CONTINUATION

Resume from this baseline

Use this PDF as the handoff document for the next LOS discussion.

Current direction to implement

Prioritize open self-registration, independent/class-linked learning, optional parent links, free dashboards, a free code-based progress tracker, the shared backend, consented summary sync and offline learning. Introduce the shared-model agent router with explicit quota handling before expanding paid inference.

Later-phase capabilities

Add on-demand personalized explanatory video generation when quality, latency and cost are practical. The ambition includes richer 3D educational visuals and personalized topic explanations. Provide teachers with additional creation/teaching capabilities in that future phase. These features are intentionally outside the current zero-budget launch.

Future development themes

Strengthen personalization, broaden offline support, retain clear privacy controls, separate public product messaging from internal design documentation, and maintain desktop/mobile feature parity. Future work must preserve access for learners without a teacher or linked parent.

Continuation instruction

When continuing, first read this PDF and treat its decisions as the baseline. Do not restart the earlier onboarding questions unless Aaryan explicitly changes a decision. Record new decisions as an updated version and distinguish actual implementation from planned behavior.

Keep fixed unless changed	Next discussion can build on
Independent access; optional adult links	Learner journeys and link verification.
Local camera processing; consented summaries	Concrete data schema, retention controls and device implementation.
Free progress and quota-limited AI	Integration tests, measured model quality and operational capacity.
Monthly credits plus optional top-ups	A costed price/allowance design when paid rollout is being prepared.

Record basis: Supplied LOS discussion excerpts. Repository: github.com/aaryankasaudhan91-dev/Learning-operating-system-LOS-. This PDF does not modify repository code.